

Industrial Internship Report on " Prediction of Agriculture Crop Production in India"

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Executive Summary

This report provides details of the Industrial Internship provided by Upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt. Ltd. (UCT).

This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time.

My project was "Prediction of Agriculture Crop Production in India Using Machine Learning ." The objective of this project was to develop a machine learning model capable of predicting agricultural crop production based on cultivation and production-related factors. The project involved data preprocessing, exploratory data analysis (EDA), outlier detection and removal using the IQR method, feature selection, model training using Linear Regression, and evaluation using Mean Absolute Error (MAE) and Mean Squared Error (MSE). The developed model helps analyze agricultural data and predict crop production more effectively.

This internship gave me a very good opportunity to get exposure to industrial problems and design/implement solutions for them. It was an overall great experience to have this internship.

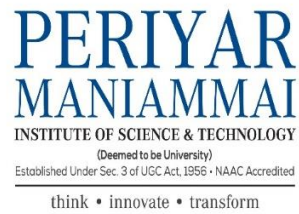


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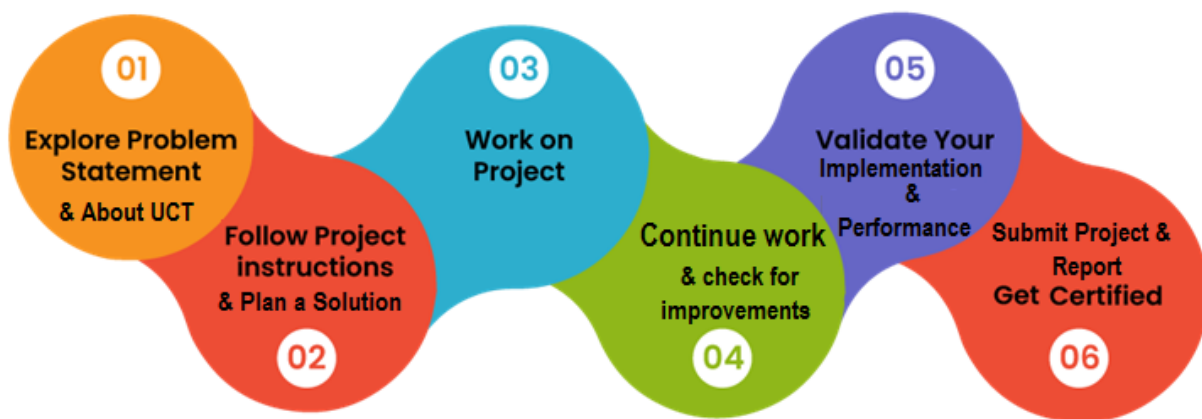
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1 Preface

The six-week Industrial Internship organized by **Upskill Campus** and **The IoT Academy** in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)** provided me with an excellent opportunity to gain practical knowledge in Data Science and Machine Learning. This internship helped me understand how theoretical concepts can be applied to solve real-world industrial problems, thereby enhancing my technical, analytical, and problem-solving skills.

As part of the internship, I worked on the project "**Prediction of Agriculture Crop Production in India Using Machine Learning.**" The project involved developing a machine learning model to predict agricultural crop production using historical data. Throughout the internship, I followed a structured approach that included understanding the problem statement, implementing the solution, validating the model, and documenting the project.



The internship was a valuable learning experience that improved my understanding of machine learning, data preprocessing, model development, and performance evaluation. I sincerely thank **Upskill Campus**, **The IoT Academy**, **UniConverge Technologies Pvt. Ltd. (UCT)**, my mentors, faculty members, and everyone who supported me throughout this internship. I encourage my juniors and peers to participate in such internship programs, as they provide practical exposure, enhance technical skills, and prepare students for future careers.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

uct
Uniconverge Technologies

- IIOT Products**
We offer product ranging from Remote IOs, Wireless IOs, LoRaWAN Sensor Nodes/ Gateways, Signal converter and IoT gateways
- IIOT Solutions**
We offer solutions like OEE, Predictive Maintenance, LoRaWAN based Remote Monitoring, IoT Platform, Business Intelligence...
- OEM Services**
We offer solutions ranging from product design to final production we handle everything for you..



i. UCT IoT Platform ()

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

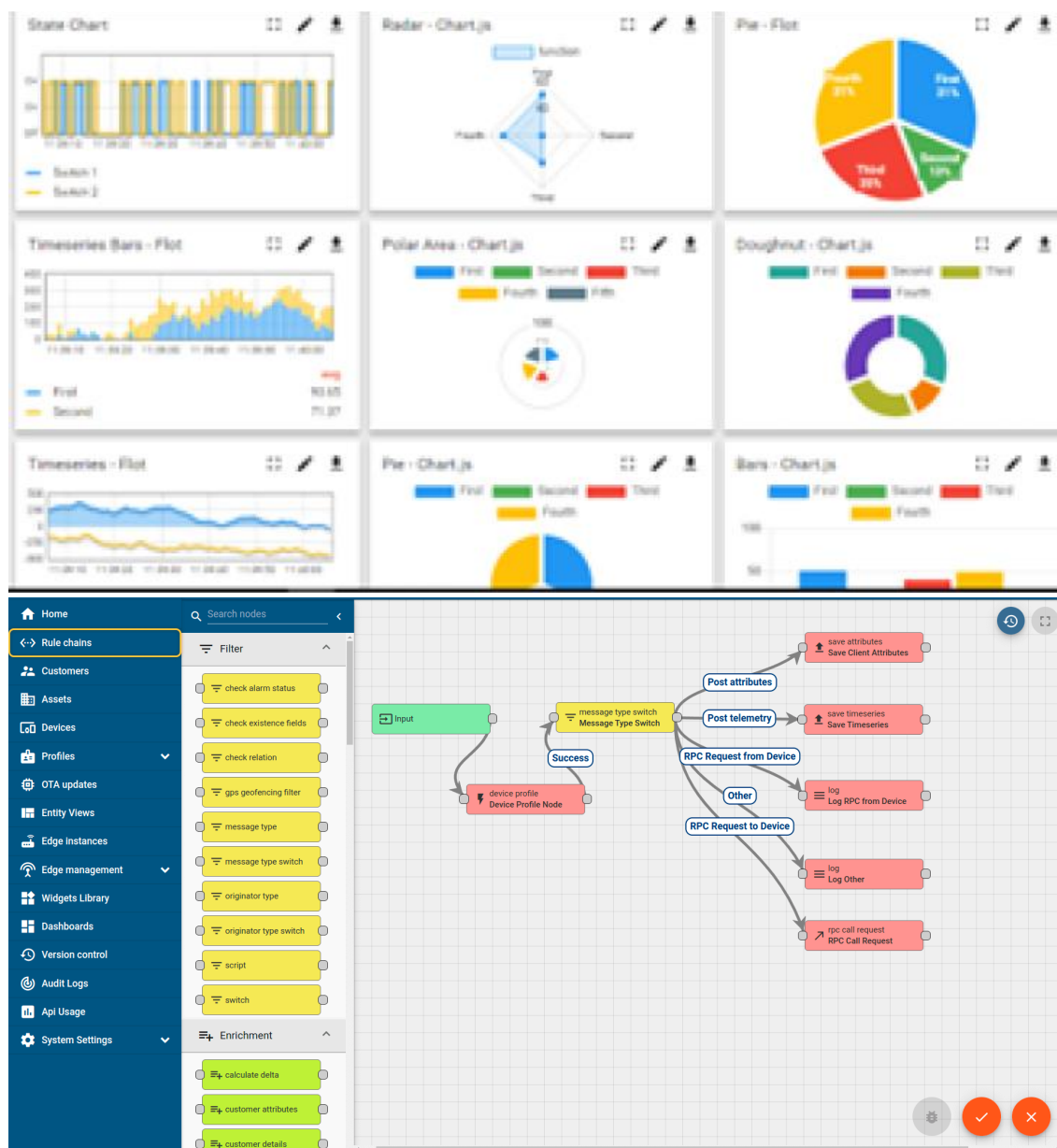
- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

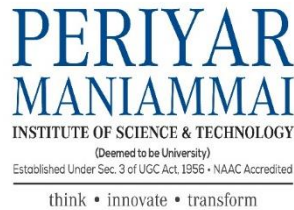
It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



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FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

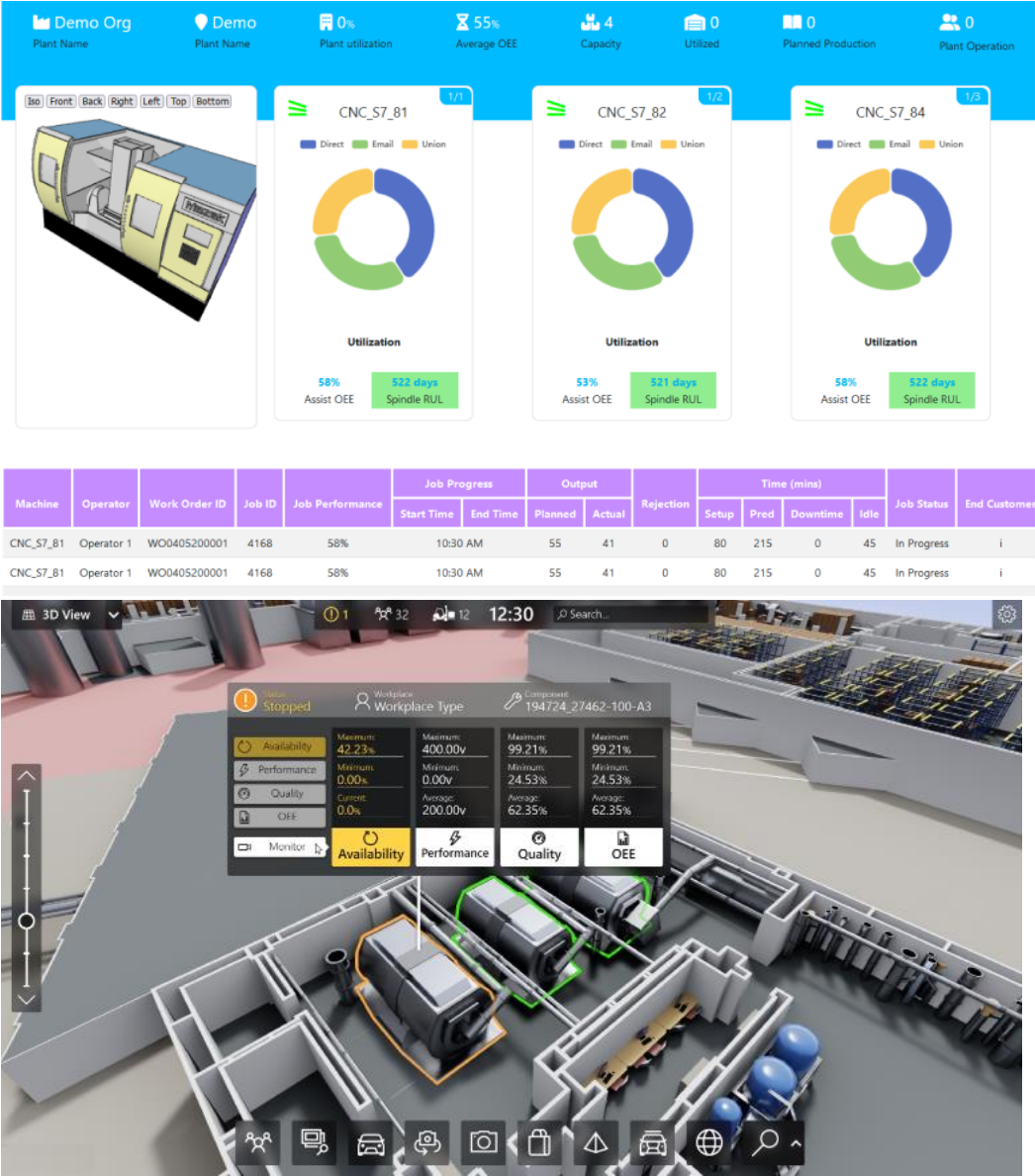
It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



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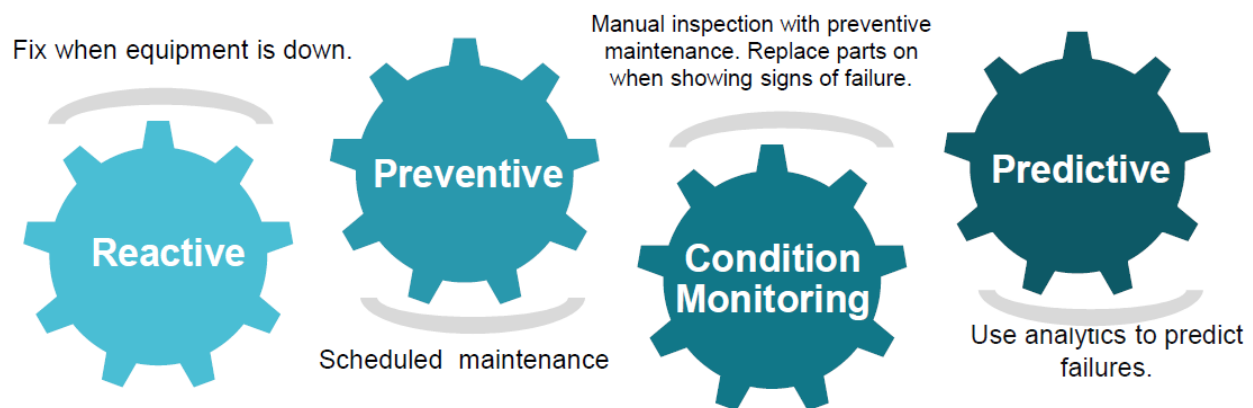


iii. based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

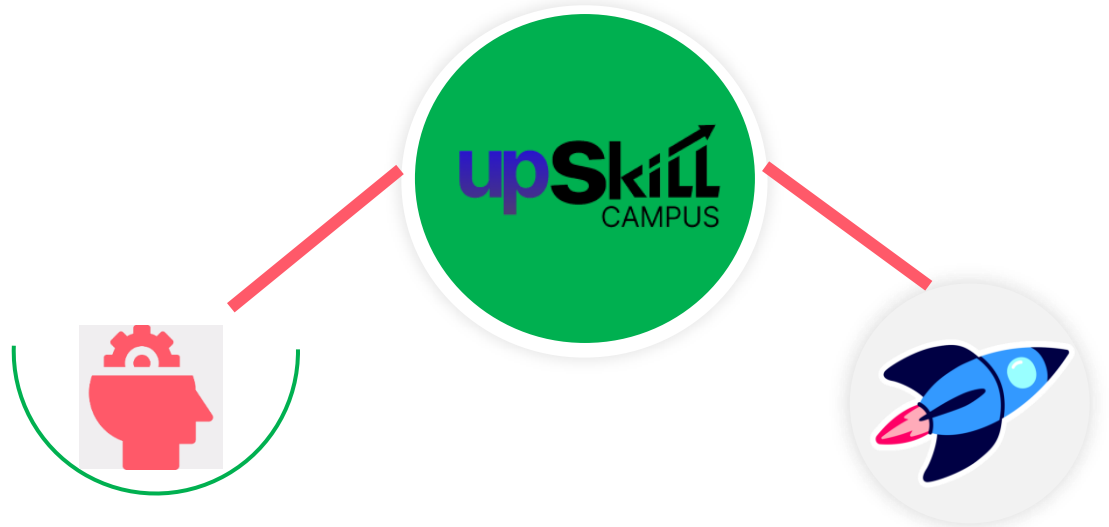
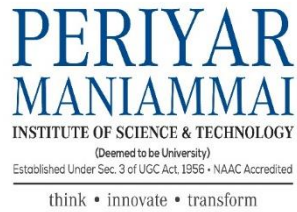
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

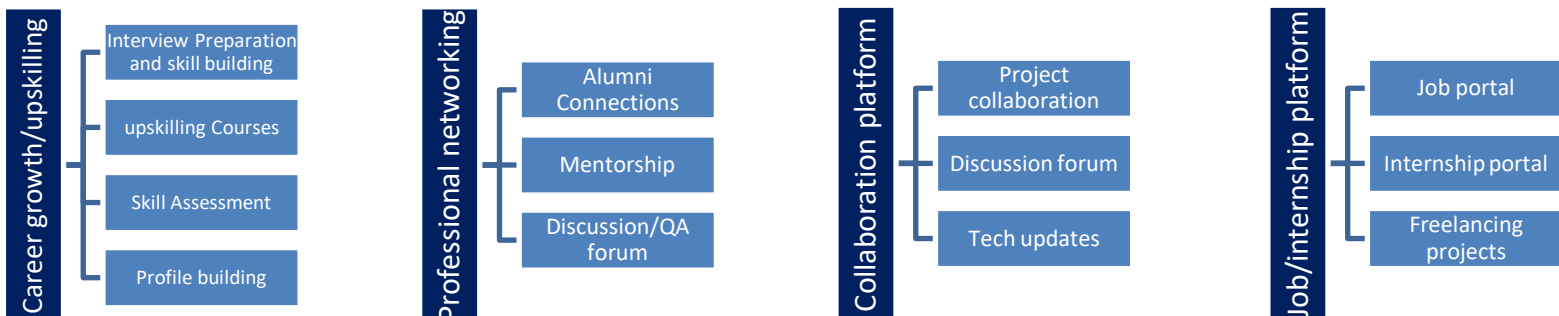
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

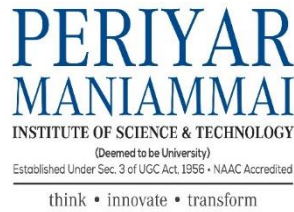


Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>





The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.3 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.4 Reference

- [1] Upskill Campus Learning Materials
- [2] The IoT Academy Learning Resources
- [3] Dataset provided by UniConverge Technologies Pvt. Ltd. (UCT)

2.5 Glossary

Terms	Acronym
AI	Artificial Intelligence
ML	Machine Learning
EDA	Exploratory Data Analysis
MAE	Mean absolute error
Linear Regression	A supervised machine learning algorithm used to predict the continuous values



3 Problem Statement

Agriculture is one of the most important sectors in India and plays a significant role in the country's economy. A large portion of the population depends on agriculture for their livelihood. However, crop production is influenced by various factors such as cultivation practices, seasonal conditions, production cost, and geographical location. Accurately predicting agricultural crop production is essential for farmers, agricultural organizations, and policymakers to make informed decisions regarding resource allocation, crop planning, and food security.

The assigned problem statement focuses on "**Prediction of Agriculture Crop Production in India Using Machine Learning.**" The objective is to develop a machine learning model that can analyze historical agricultural data and predict crop production based on the available features. Such predictions can assist in improving agricultural planning, reducing uncertainty, and supporting better decision-making in the agricultural sector.

In this project, a dataset containing agricultural production information was analyzed and preprocessed using Python. Data preprocessing techniques, including handling missing values, exploratory data analysis (EDA), and outlier removal using the Interquartile Range (IQR) method, were performed before building the prediction model. A **Linear Regression** algorithm was then implemented to predict crop production, and the model's performance was evaluated using **Mean Absolute Error (MAE)** and **Mean Squared Error (MSE)**.



4 Existing and Proposed solution

4.1 Existing Solution

Agricultural crop production is traditionally estimated using historical records, manual analysis, and expert knowledge. Farmers and agricultural organizations often rely on previous years' production data, seasonal observations, and experience to make cultivation decisions. While these methods are useful, they are time-consuming and may not accurately capture the relationships between multiple factors affecting crop production. Conventional approaches also struggle to analyze large volumes of agricultural data efficiently and may lead to less accurate predictions.

- **Limitations of Existing Solutions**
- Depend heavily on manual analysis and expert experience.
- Time-consuming when processing large datasets.
- Limited ability to identify hidden patterns in agricultural data.
- Prediction accuracy may be affected by multiple influencing factors.
- Difficult to scale for large and diverse agricultural datasets.

4.2 Proposed Solution

The proposed solution is a **Machine Learning-based crop production prediction system** developed using the **Linear Regression** algorithm. The agricultural dataset is first preprocessed by checking for missing values, performing Exploratory Data Analysis (EDA), and removing outliers using the Interquartile Range (IQR) method. After preprocessing, the dataset is divided into training and testing sets, and a Linear Regression model is trained to predict crop production. The model is evaluated using **Mean Absolute Error (MAE)** and **Mean Squared Error (MSE)** to assess its performance.

- Value Addition
- Automates the crop production prediction process.



- Provides faster and more consistent predictions than manual methods.
- Uses data preprocessing techniques to improve data quality.
- Helps identify patterns and relationships within agricultural data.
- Supports data-driven decision-making for agricultural planning and production management.

4.1 **Code submission :** <https://github.com/joshikacrisane05-tech/upskillcampus/blob/main/CropYieldPrediction.ipynb>

4.2 **Report submission :** https://github.com/joshikacrisane05-tech/upskillcampus/blob/main/CropYieldPrediction_Joshika_USC_UCT.pdf

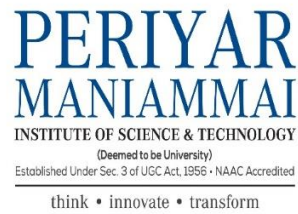
5 Proposed Design/ Model

Proposed Design

The proposed system is a Machine Learning-based model developed to predict agricultural crop production using historical agricultural data. The system follows a structured workflow that begins with data collection and preprocessing, followed by exploratory data analysis, model training, prediction, and performance evaluation. Python and its Machine Learning libraries were used to implement the complete solution.

The workflow of the proposed system consists of the following stages:

- Dataset Collection
- Data Preprocessing
- Exploratory Data Analysis (EDA)
- Outlier Detection and Removal (IQR Method)



- Feature Selection
- Train-Test Split
- Model Training using Linear Regression
- Prediction
- Model Evaluation using MAE and MSE
- Result Visualization

5.1 High Level Diagram (if applicable)

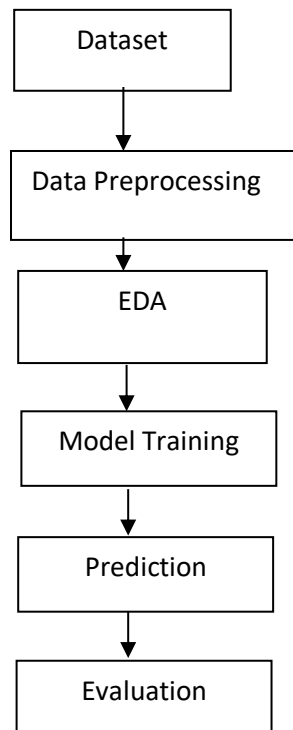
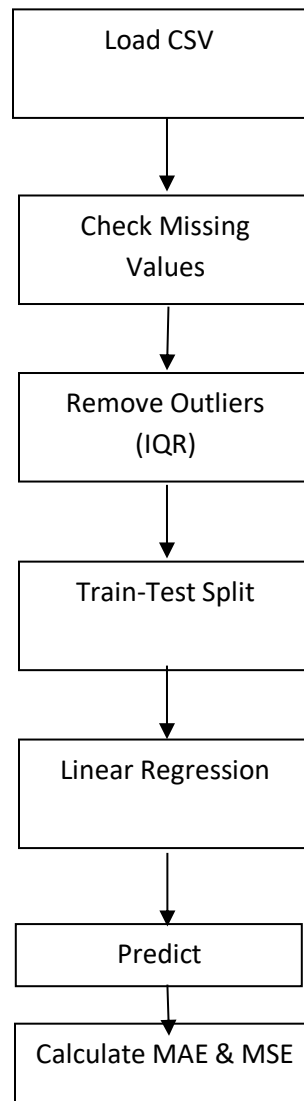




Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)





6 Performance Test

The developed machine learning model was evaluated to determine its effectiveness in predicting agricultural crop production. Before training the model, the dataset was preprocessed by handling missing values, performing exploratory data analysis (EDA), and removing outliers using the Interquartile Range (IQR) method. These preprocessing steps improved the quality of the dataset and helped the model learn meaningful patterns.

The performance of the Linear Regression model was evaluated using **Mean Absolute Error (MAE)** and **Mean Squared Error (MSE)**. These metrics were used to measure the difference between the actual crop production values and the predicted values. The prediction results were also visualized using an Actual vs Predicted graph to assess the model's performance.

The model's performance depends on factors such as dataset quality, availability of relevant features, and the presence of outliers. Proper preprocessing and feature selection contributed to improving the prediction accuracy of the model.

6.1 Test Plan/ Test Cases

Test Case	Expected Result	Status
Load the agricultural dataset	Dataset loaded successfully	Passed
Check for missing values	Missing values identified and handled	Passed
Perform data preprocessing	Dataset cleaned successfully	Passed
Remove outliers using IQR	Outliers removed from the dataset	Passed
Train the Linear Regression model	Model trained successfully	Passed
Predict crop production	Predicted values generated	Passed

Test Case	Expected Result	Status
Evaluate the model using MAE and MSE	Performance metrics calculated successfully	Passed



6.2 Test Procedure

1. Load the agricultural dataset into Google Colaboratory.
2. Check the dataset for missing values and duplicates.
3. Perform exploratory data analysis (EDA).
4. Detect and remove outliers using the IQR method.
5. Select the required features and target variable.
6. Split the dataset into training and testing sets.
7. Train the Linear Regression model using the training data.
8. Generate predictions using the testing data.
9. Evaluate the model using MAE and MSE.
10. Visualize the prediction results using graphs.

6.3 Performance Outcome

The Linear Regression model was successfully trained using the preprocessed agricultural dataset. The model generated crop production predictions based on the selected input features. Performance evaluation using **Mean Absolute Error (MAE)** and **Mean Squared Error (MSE)** indicated that the model was able to learn the relationship between the input features and crop production effectively. The Actual vs Predicted graph further demonstrated the model's prediction capability. The project successfully achieved its objective of developing a machine learning model for agricultural crop production prediction.

7 My learnings

The six-week Industrial Internship provided me with valuable practical experience in Data Science and Machine Learning. During the internship, I learned how to approach a real-world problem systematically, starting from understanding the problem statement to developing and evaluating a machine learning model.

Through this project, I gained hands-on experience in data preprocessing, exploratory data analysis (EDA), feature selection, outlier detection using the Interquartile Range (IQR) method, and implementing



the Linear Regression algorithm. I also improved my programming skills using Python and became familiar with libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn. In addition, I learned how to evaluate machine learning models using performance metrics such as Mean Absolute Error (MAE) and Mean Squared Error (MSE).

This internship strengthened my analytical thinking, problem-solving abilities, and understanding of the complete machine learning workflow. The knowledge and practical experience gained during this internship have increased my confidence in working on real-world data science projects and will be highly beneficial for my future career in Machine Learning and Artificial Intelligence.

8 Future work scope

The developed machine learning model successfully predicts agricultural crop production using historical agricultural data. However, there is scope for further improvement and enhancement in the future.

Future work may include using larger and more diverse datasets to improve the model's accuracy and generalization. Additional factors such as weather conditions, rainfall, soil quality, temperature, humidity, and fertilizer usage can be incorporated to build a more comprehensive prediction model. Advanced machine learning algorithms such as **Random Forest**, **Decision Tree**, **XGBoost**, or **Artificial Neural Networks** can also be explored and compared with Linear Regression to achieve better prediction performance.

Furthermore, the model can be integrated into a web or mobile application to provide farmers and agricultural organizations with real-time crop production predictions. Such enhancements would make the system more practical, scalable, and beneficial for supporting data-driven decision-making in the agricultural sector.

